

Modified De-interlacing method based on edge direction

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Abstract

This paper proposes a de-interlacing method which can detect correct edge directions for interpolation. In the traditional method, the determination of the edge direction is done by only the 2 pixels in that direction. This method sometimes chooses incorrect edge direction. The proposed method adopts 6 pixels in the corresponding direction and an average value is calculated. Since more information is provided, the correct edge direction can be allocated. And as a consequence, sawtooth effects and other artifacts can be removed.

1 Introduction

Interlaced video sequence consists of even fields and odd fields. The even field contains the signal of even-numbered lines of a frame while the odd field contains odd lines. For a video sequence, the two fields alternate in time.

Interlaced scan technique produces defects, such as edge flicker and line crawling which cause visual artifacts. These artifacts are most noticeable when horizontal movement exists between the two frames. De-interlacing technique is proposed to solve above problems [1-3]. Such techniques essentially convert interlaced video sequence into progressive ones and have been widely used to reduce the visual artifacts.

Two basic de-interlacing techniques are combination and extension [4]. Weaving is the simplest combination method which combines the even and odd fields into a frame. But, this method causes line crawling effects in the motion region. Also, although this technique retains full vertical resolution, it can only obtain half of the temporal resolution. The other technique BOB is an extension method. It takes the lines of either even or odd field and duplicates each line to form an entire frame. However, it prevents the line crawling effects, but it causes a noticeable reduction in picture quality.

The edge-based line average (ELA) algorithm is a well-known interpolation method in the spatial domain, which also belongs to extension method.

This method performs interpolation in the directional correlation among pixels, and it only requires little computational load. Based on the ELA algorithm, several approaches are proposed [5-8]. However, the accuracy of the ELA algorithm which detects the edge direction by one difference among two pixels is not better enough.

In this paper, we propose a modified ELA method for de-interlacing. The proposed method uses three differences in each direction to estimate the edge direction for interpolating. Since the edge directions are detected more accurately, the de-interlaced frame has better visual effect and higher image quality.

This paper is organized as follows. In the next section, review of de-interlacing is presented briefly. In section 3, the proposed method is described in detail. Section 4 presents and discusses the experimental results. Finally, the conclusions are given in section 5.

2 De-interlacing methods

De-interlacing is a frame format conversion from interlaced to progressive frames. The original video sequence consists of even fields and odd fields. In this paper, we reserve the even field and operate on it to obtain the odd field to form an entire frame.

Let the reserved even field and the reconstructed frame be denoted by g and f , respectively. First, we assume the entire frame f is undefined. Let the size of the frame $f(i, j)$ be $0 \leq i < M$, $0 \leq j < N$, and $g(i, j)$ be $0 \leq i < M$, $0 \leq j < N/2$. First, we copy the lines of g to the even-numbered scan lines in the entire frame as shown in Fig.1. This step can be represented by the following equation

$$f(i, 2k) = g(i, k) \quad (1)$$

where $0 \leq i < M$ and $0 \leq k < N/2$. Similarly, the odd field signal can be represented as $f(i, 2k+1)$ which is not defined.

In obtaining the odd field lines of f , BOB is the simplest method which copies even field signal to odd field. The equation is represented by

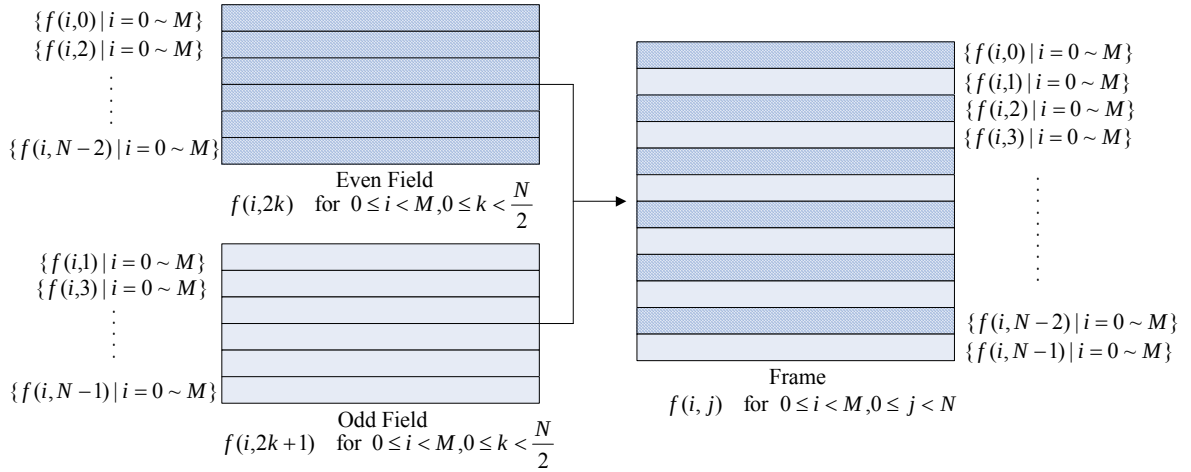


Fig.1. The relationship between even field , odd field, and entire field frame

$$f(i,2k+1) = f(i,2k) \quad (2)$$

This method could cost less, but the results are unacceptable due to the edge sawtooth effects.

Linear interpolation is the most commonly used method for de-interlacing. Each pixel in the odd field $f(i,2k+1)$, which is centered on two adjacent even-numbered lines, is estimated by averaging the upper and lower pixel values. The interpolation equation is given as

$$f(i,2k+1) = (f(i,2k) + f(i,2(k+1))) / 2 \quad (3)$$

This method solves the de-interlacing problem by intuition, and it is easy to implement. But, some sawtooth effect still exists in some regions. Also, this method suffers the existence of edge blurring effect in the de-interlaced frame.

The edge-based line average (ELA) algorithm utilizes directional correlation among pixels to perform linear interpolation. There are three detection directions as shown in Fig.2, which are vertical, diagonal and sub-diagonal. In each direction, the difference is calculated. The interpolation is performed using averaging according to the direction with minimal difference among these three directions.

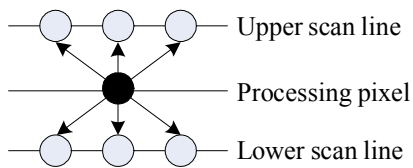


Fig.2. ELA algorithm

Assume the processing pixel is represented by

$f(i,2k+1)$, the difference of each of the three directions in the ELA algorithm is calculated by

$$e_1 = |(f(i-1,2k) - f(i+1,2(k+1)))| \quad (4)$$

$$e_2 = |(f(i,2k) - f(i,2(k+1)))| \quad (5)$$

$$e_3 = |(f(i+1,2k) - f(i-1,2(k+1)))| \quad (6)$$

In terms of filter, the algorithm can be re-expressed as given in Fig.3.

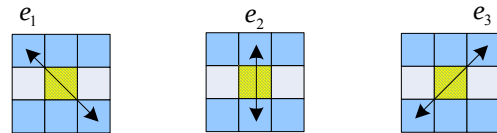


Fig.3. The three filters for ELA algorithm

As the differences are calculated, the direction p defined by

$$p = \arg(\min(e_1, e_2, e_3)) \quad (7)$$

is the best direction detected. The pixel value is then the average of the two opposite pixels in the p direction. This method outperforms the previous methods. However, some of the directions are incorrectly determined.

3 Proposed method

The proposed de-interlacing method aims at the reduction of the edge sawtooth and the elimination of the edge blurring effect after the linear interpolation. In this method, the difference to ELA algorithm is the number of differences in

each edge direction. There are three differences in one direction in order to avoid the noise near the edge so as to allocate the correct direction.

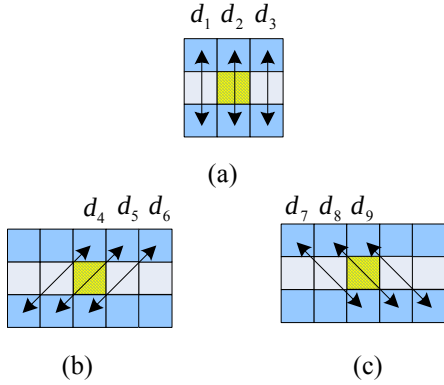


Fig.4. The three filters for proposed method

The three differences are shown in Fig.4. The center cross pixel is the one to be processed. The main steps of this proposed method are given as following:

- Step 1: Utilize the three differences in Fig.4. to detect which edge direction passes through the pixel to be processed.
- Step 2: Execute the linear interpolation by the edge direction of Step 1 to obtain the processed pixel value.

In Step 1, for one direction, the differences d_1, d_2, d_3 are calculated between two opposite pixels indicated by an arrow with following equation

$$d_1 = |(f(i-1, 2k) - f(i-1, 2(k+1)))| \quad (8)$$

$$d_2 = |(f(i, 2k) - f(i, 2(k+1)))| \quad (9)$$

$$d_3 = |(f(i+1, 2k) - f(i+1, 2(k+1)))| \quad (10)$$

Then, the feature value v_a of the filter of Fig.4(a) is determined by averaging the three differences d_1, d_2, d_3 .

$$v_a = (d_1 + d_2 + d_3) / 3 \quad (11)$$

In the same manner, the feature values v_b and v_c of Fig.4(b) and Fig.4(c) are calculated similar to above equations. According to the minimum of the feature values v_a, v_b, v_c , an edge direction can be detected.

In Step 2, the value of the pixel to be estimated is calculated by interpolating the two opposite pixels with the middle arrow in the edge direction detected in Step 1.

4 Experimental results

In this section, the image quality and the edge smoothness of the images and subimages are compared of the proposed method to the other methods. In the experiments, we take out the even field signal from an entire frame, and then utilize it to calculate the pixel values in the odd field. The methods compared are BOB de-interlacing, linear interpolation, ELA algorithm, and the proposed method. The results of the de-interlaced frames are shown in Fig.4, 6, and 8, and the subimages enlarged for advance observation of the edge sawtooth effects are shown in Fig.5, 7, and 9.

The six figures have similar results. That is, BOB de-interlacing and linear interpolation can not solve the edge sawtooth effect. Also, the proposed method is better than ELA algorithm in visual effect. The Peak Signal to Noise Ratio (PSNR) of proposed method compared to ELA algorithm increases from 0.5 to 2 dB. The detail comparison of ELA algorithm and the proposed method as discussed as follows.

In Fig.5, the subimage is picked on the fringe of hat slanted to the left. The lower slope is dealt nice for both ELA algorithm and the proposed method. But, the upper slope has edge sawtooth in ELA algorithm, and the proposed method has smooth edge.

In Fig.7, the subimage is the wing with the words "F-16." The edges in the subimage are smooth for ELA algorithm and the proposed method. However, there are noises connecting the two digits "1" and "6" in ELA algorithm due to the incorrect selection of the edge directions. While in our method, the two digits can be clearly separated.

In Fig.9, the subimage contains complicated lines is one frame of video sequence. The approached edge lines would influence the determination of ELA algorithm and the proposed method. Since the ELA algorithm only references one difference in each direction, the experimental results are unsatisfied in comparison to our method.

5 Conclusion

In this paper, the proposed method utilizes the average of three differences in each of the three angles to determine the edge direction for de-interlacing. Experimental results show that the proposed method obtains better image quality with 0.5 to 2 dB better than that of the ELA algorithm. Also, our method can successfully reduce the edge sawtooth effect.

References

- [1] G. de Haan and E. B. Bellers, "De-interlacing: an overview," *Proc. of the IEEE*, vol. 86, no. 9, pp. 1839-1857, 1998.
- [2] G. D. Haan and E. B. Bellers, "De-interlacing of video data," *IEEE Trans. Consumer Electron.*, vol. 43, no. 3, pp. 819-825, 1997.
- [3] Y. Shimodaira, "Fundamental phenomena underlying artifacts induced by image motion and the solutions for decreasing the artifacts on FPDs," *SID 03 DIGEST*, pp. 1034-1037, 2003.
- [4] <http://en.wikipedia.org/wiki/Deinterlacing>
- [5] T. Doyle, "Interlaced to Sequential Conversion for EDTV Applications," in *Proc. 2nd Int. Workshop Signal Processing of HDTV*, pp.412- 430, 1998.
- [6] M.H. Lee, J.H. Kim, J.S. Lee, K.K. Ryu, and D.I. Song, "A New Algorithm for Interlaced to Progressive Scan Conversion Based on Directional Correlations and its IC Design," *IEEE Trans. Consumer Electron*, vol. 40, no. 2, pp. 119-129, May. 1994.
- [7] C.J. Kuo, C. Liao, and C.C. Lin, "Adaptive Interpolation Technique for Scanning Rate Conversion," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 6, no. 3, pp. 317-321, 1996.
- [8] T. Chen, H. R. Wu, and Z. H. Yu, "An efficient edge line average interpolation algorithm for de-interlacing," *Visual Communications and Image Processing 2000, Proc. of SPIE*, pp. 1551-1558, 2000

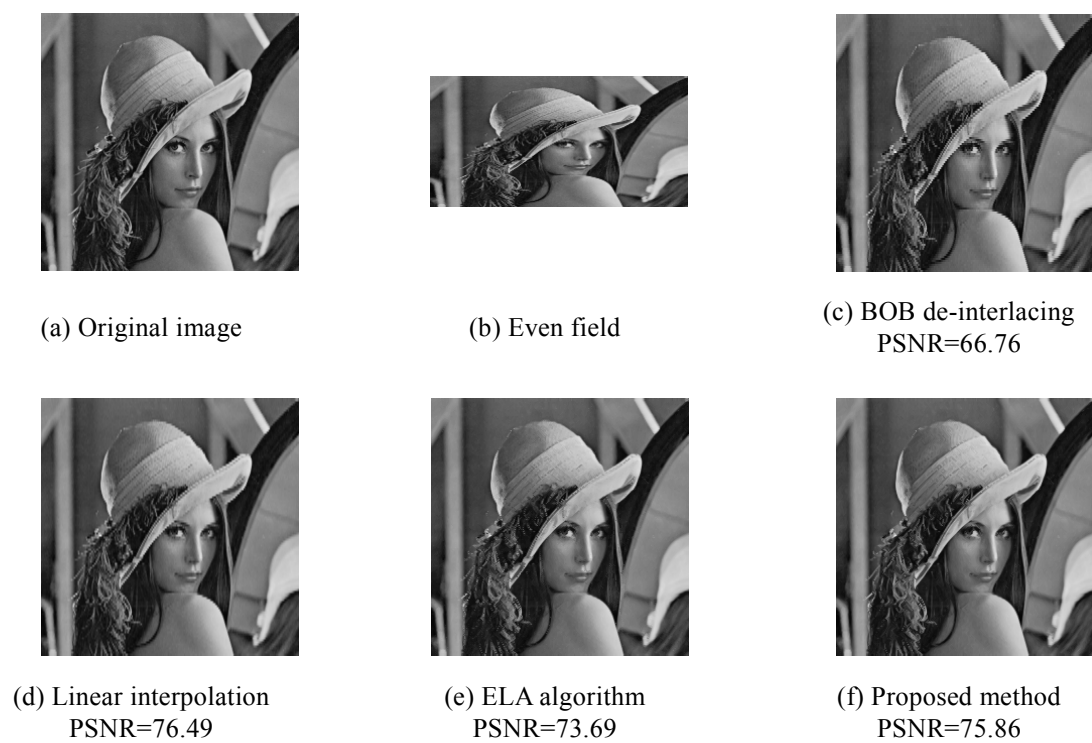


Fig.4. The results of Lena image for de-interlacing (reduce 50%)

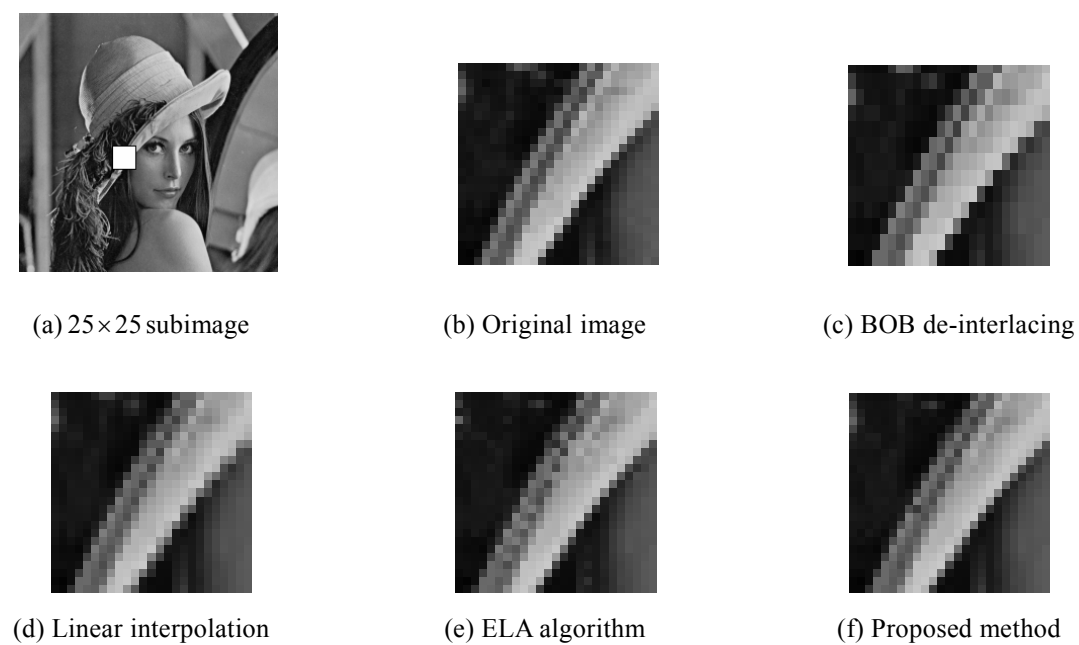


Fig.5. The subimages of de-interlaced Lena images (enlarge 400%)



(a) Original image



(b) Even field



(c) BOB de-interlacing
PSNR=55.57



(d) Linear interpolation
PSNR=62.23



(e) ELA algorithm
PSNR=60.07

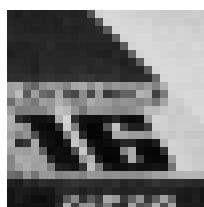


(f) Proposed method
PSNR=60.99

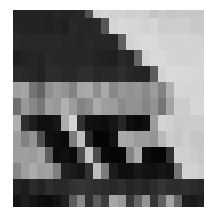
Fig.6. The results of F-16 image for de-interlacing (reduce 50%)



(a) 25×25 subimage



(b) Original image



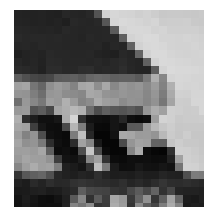
(c) BOB de-interlacing



(d) Linear interpolation



(e) ELA algorithm



(f) Proposed method

Fig.7. The subimages of de-interlaced F-16 images (enlarge 400%)

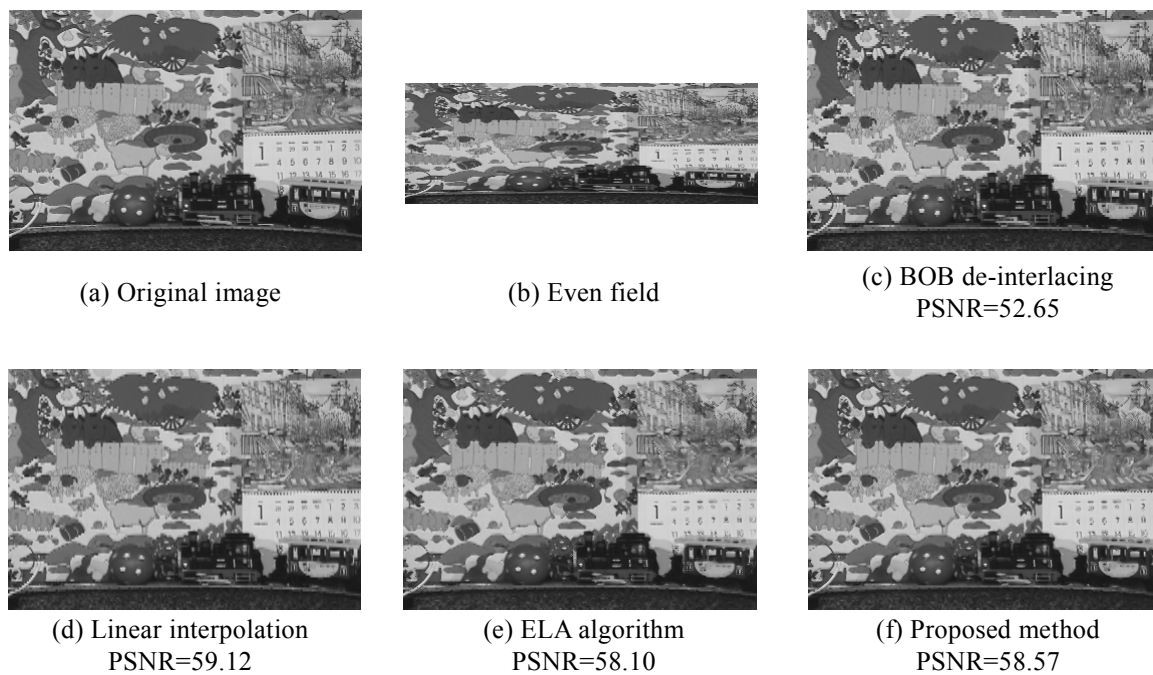


Fig.8. The results of Calendar image for de-interlacing (reduce 50%)

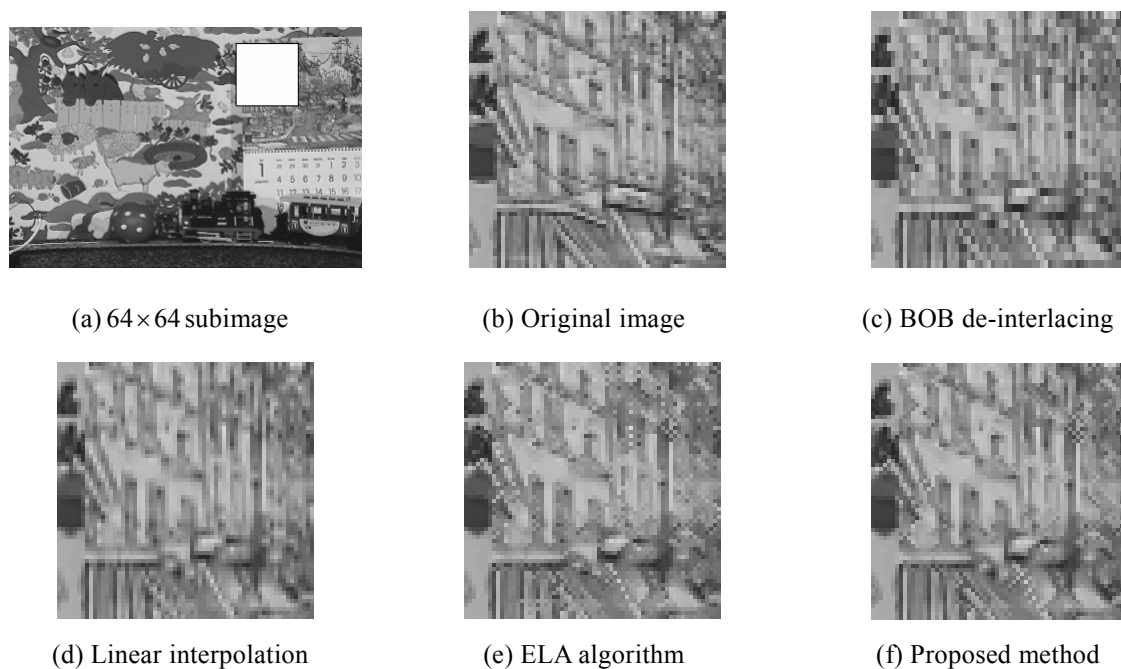


Fig.9. The subimages of de-interlaced Calendar images (enlarge 200%)